DESKTOP PLATFORM GUIDE · VERSION 1.0.0 · 2026-11-06

adl Desktop Apps Guidelines

UI, UX and platform rules for adl's JavaFX desktop apps. Editora is the reference implementation; every rule names the class or file to copy from.

Editora · reference | Torrenta | Termina | Modula | Mirada | nux

MUST release blocker for a new app SHOULD expected unless the app says why not

MAY copy when the app grows into it

CONTENTS

Principles

New app checklist

Platform baseline

Repository documents

Code quality gates

Keyboard first

Window shell

Settings

Diagnostics and support

Command line and launch

Internationalization

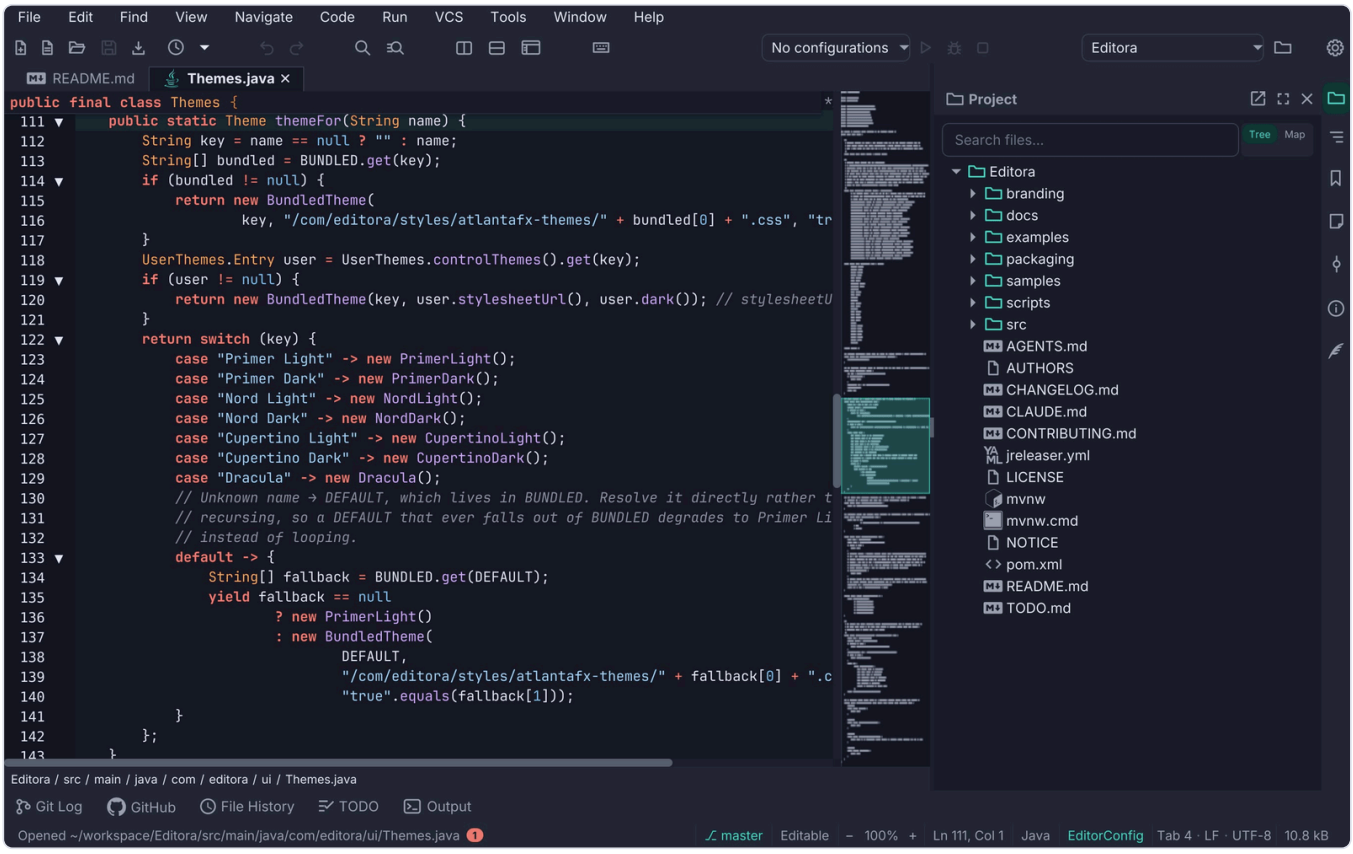
Packaging, signing and release

Feature matrix

Shared features

Gaps per app

Principles



Editora, the reference implementation: one window where every action is a command, reachable from a keymap or the palette.

1. **Keyboard first.** Every action works without a mouse. The mouse is a shortcut to commands, not the other way round.
2. **Every feature is a command.** One registry of commands with ids. Menus, toolbar buttons, the command palette and key bindings all run commands by id. Nothing is wired straight to logic.
3. **Changes apply as they are made.** Settings have no Save button. A change reaches every open window.
4. **Nothing fails silently.** Every failure reaches the status bar, the Message Log or the Debug Log, and the user can turn any of it into a GitHub issue.
5. **Local first, nothing sent without asking.** No telemetry. The only automatic network call is a once-a-day release check the user can turn off. Problem reports are built locally and submitted by the user.
6. **Same app on every OS.** Windows, macOS and Linux get the same features and the same installer quality. Platform differences live behind one class, never scattered `if (isMac)` checks.
7. **Users' data outlives versions.** Config files carry a schema version and migrate forward. An older build never overwrites a newer config.

New app checklist

Each group links to the page with the rules and the Editora code to copy. Everything here is a MUST unless it says SHOULD.

Platform

Platform baseline

- [ ] JDK 25 or newer (`release` 25 is the floor) and JavaFX 27 or newer
- [ ] One JPMS module, `com.<app>`
- [ ] Maven Wrapper committed (`mvnw` , `mvnw.cmd` , `.mvn/wrapper/`)
- [ ] AtlantaFX light and dark themes, following the system by default
- [ ] Inter bundled and attached to every scene and dialog (SHOULD)

Code quality

Code quality gates

- [ ] Every package `@NullMarked` ; NullAway on Error Prone fails the build
- [ ] Spotless with Palantir Java Format
- [ ] JUnit 5 and headless FX tests (`-Dglass.platform=Headless`), FX tests tagged `fx`
- [ ] JaCoCo check at 0.90 lines and 0.90 methods

Keyboard and commands

Keyboard first

- [ ] One `CommandRegistry` : every action is a command with an id
- [ ] Command Palette as an in-scene overlay, not a `Popup`
- [ ] Emacs and CUA keymaps at least, with user overrides on top

Settings

Settings

- [ ] Settings window with a page sidebar, search and live apply
- [ ] A palette command for every setting
- [ ] Versioned config in `~/.<app>` , with migrations and downgrade backups
- [ ] Export and import of the config folder
- [ ] Settings sync through a Git/GitHub repository (SHOULD)

Diagnostics

Diagnostics and support

- [ ] Status-bar echo and Message Log
- [ ] Debug Log with a session file, and an uncaught exception handler
- [ ] Daily update check with an off switch; About shows versions and update status
- [ ] Report a Problem (redacted, prefilled GitHub issue) and `--report`

Command line and languages

Command line · Internationalization

- [ ] `--help`, `--version`, `--config-dir`, `--dev`
- [ ] Single instance per config folder
- [ ] At least six message catalogs, with a parity test

Repository

Repository documents

- [ ] `README.md` with the badge row and a light/dark hero screenshot
- [ ] `CHANGELOG.md` in Keep a Changelog format
- [ ] MIT `LICENSE`, and `NOTICE` with every third-party licence
- [ ] `AGENTS.md`, with a one-line `CLAUDE.md` pointing to it
- [ ] `CONTRIBUTING.md`, `docs/` and `branding/`
- [ ] `.github/ISSUE_TEMPLATE/bug_report.yml`

Packaging and signing

Packaging, signing and release

- [ ] jpackage installers through JReleaser: Linux x64 and arm64, macOS arm64 and x64, Windows x64
- [ ] macOS `.dmg` signed with Developer ID, notarized and stapled
- [ ] Linux downloads with `.asc` signatures from the shared release key
- [ ] Build provenance attestations for every download
- [ ] Windows `.msi` Authenticode, once the certificate is chosen (SHOULD)
- [ ] Homebrew cask in `adriandeleon/homebrew-tap` and a winget package `AdrianDeLeon.<App>` (SHOULD)

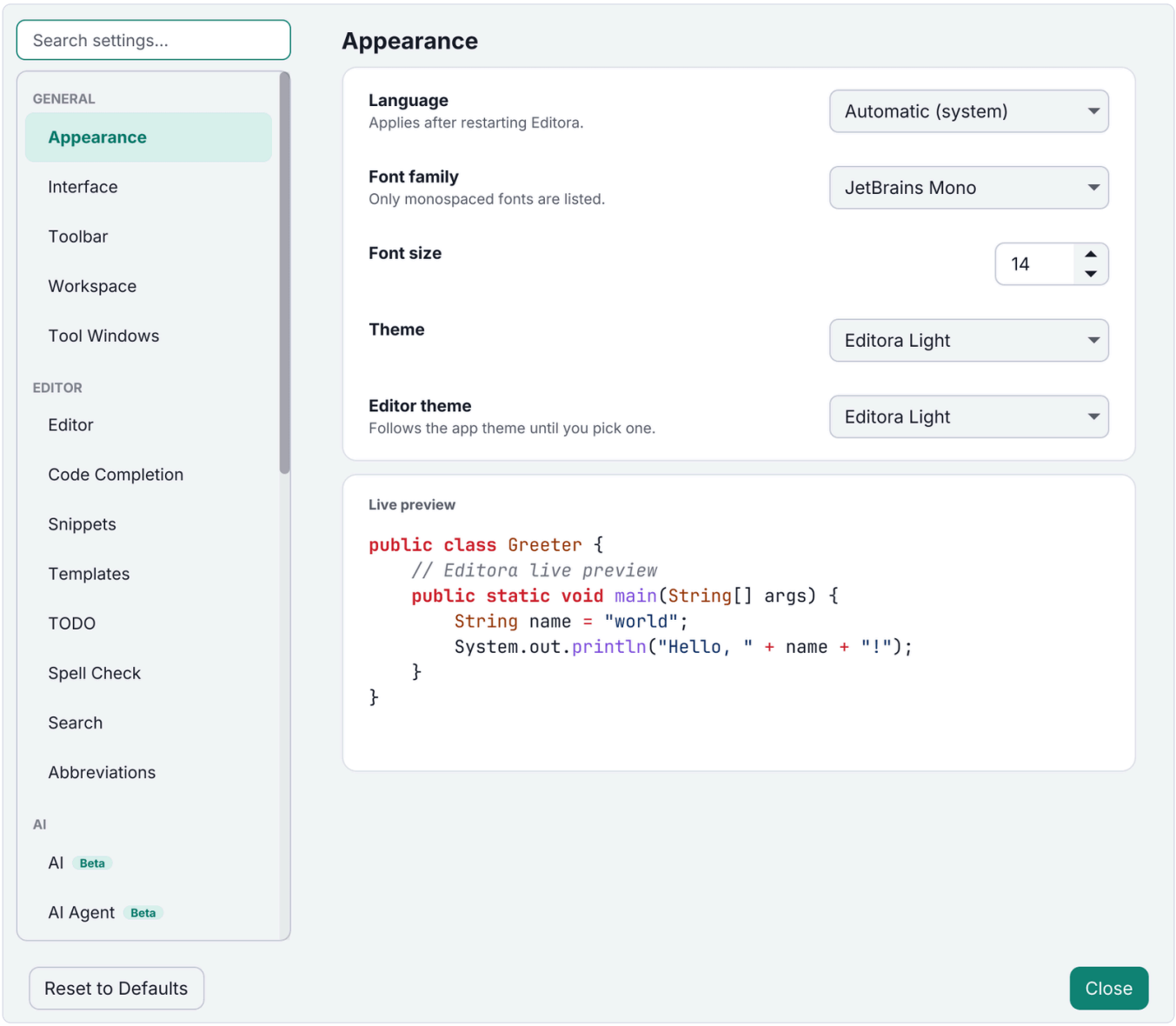
CI/CD (GitHub Actions)

CI/CD

- [ ] `ci.yml` on every push and pull request: Format and Build & Test, both required checks for `master`
- [ ] `full-tests.yml` on Linux, macOS and Windows: daily, on a `full-ci` label, before every release (SHOULD)
- [ ] `release.yml` on a `v*` tag that matches the pom, with a `workflow_dispatch` dry run
- [ ] Every third-party action pinned to a commit SHA
- [ ] Dependabot for `github-actions` and `maven`
- [ ] `permissions: contents: read` by default; write only in publishing jobs
- [ ] `timeout-minutes` on every job; superseded pull-request runs cancelled
- [ ] Test and coverage reports uploaded; coverage totals in the job summary
- [ ] Signing skipped without secrets, but a tag without them fails
- [ ] A smoke launch of the packaged app on each OS before release

Platform baseline

Area	Rule	Level	Editora reference
JDK	**JDK 25 or newer.** `maven.compiler.release` 25 is the floor, not the ceiling: build and bundle a current JDK (the enforcer range is `[25,)`, open-ended). The bundled runtime is jlinked; users never install Java	MUST	`pom.xml`
JavaFX	**JavaFX 27 or newer** (`javafx.version` 27 is the floor)	MUST	`pom.xml`
Modules	One named JPMS module (`com.<app>`); `module-info.java` opens only what reflection, FXML and Jackson need	SHOULD	`src/main/java/module-info.java`
Theme toolkit	**AtlantaFX 3.x** (`atlantafx-base`) for controls; light and dark themes; app CSS uses AtlantaFX `-color-*` tokens, never hex literals	MUST	`ui/Themes.java`, `app.css`
UI font	Bundle **Inter** (Regular, Bold, Italic, BoldItalic + `OFL.txt`); register with `Font.loadFont` before any stylesheet; attach `ui-font.css` to every scene and dialog pane. Gives the family one look on every OS	SHOULD	`ui/Fonts.java`
Build	**Maven** with the **Maven Wrapper** (`mvnw`, `mvnw.cmd`) committed, so `./mvnw` works on a clean machine	MUST	`mvnw`, `.mvn/wrapper/`
Null safety	**JSpecify** annotations, checked by **NullAway** on **Error Prone** at compile time. Every package is `@NullMarked` in its own `package-info.java`	MUST	`docs/conventions.md#null-annotations-jspecify-checked-by-nullaway`
Formatting	**Spotless** with Palantir Java Format; CI fails an unformatted change	MUST	`pom.xml` (spotless plugin)
Tests	**JUnit 5** + **TestFX** over JavaFX's built-in **Headless** glass platform (`-Dglass.platform=Headless`), so FX tests need no display, no xvfb and no Monocle	MUST	`docs/testing.md`, `FxTestSupport`
Coverage	**JaCoCo** `check` at `mvn verify`: **≥ 0.90 lines, ≥ 0.90 methods**, ≥ 0.79 branches, plus per-package floors	MUST	`pom.xml` `jacoco-check`
Logging	`java.util.logging`, captured in-app by the Debug Log	SHOULD	`ui/DebugLog.java`
JSON config	Jackson POJOs with a `SCHEMA_VERSION` each	SHOULD	`config/Settings`, `config/migration/ConfigSchema.java`



Settings → Appearance in Editora: language, fonts, app theme and editor theme, with a live preview.

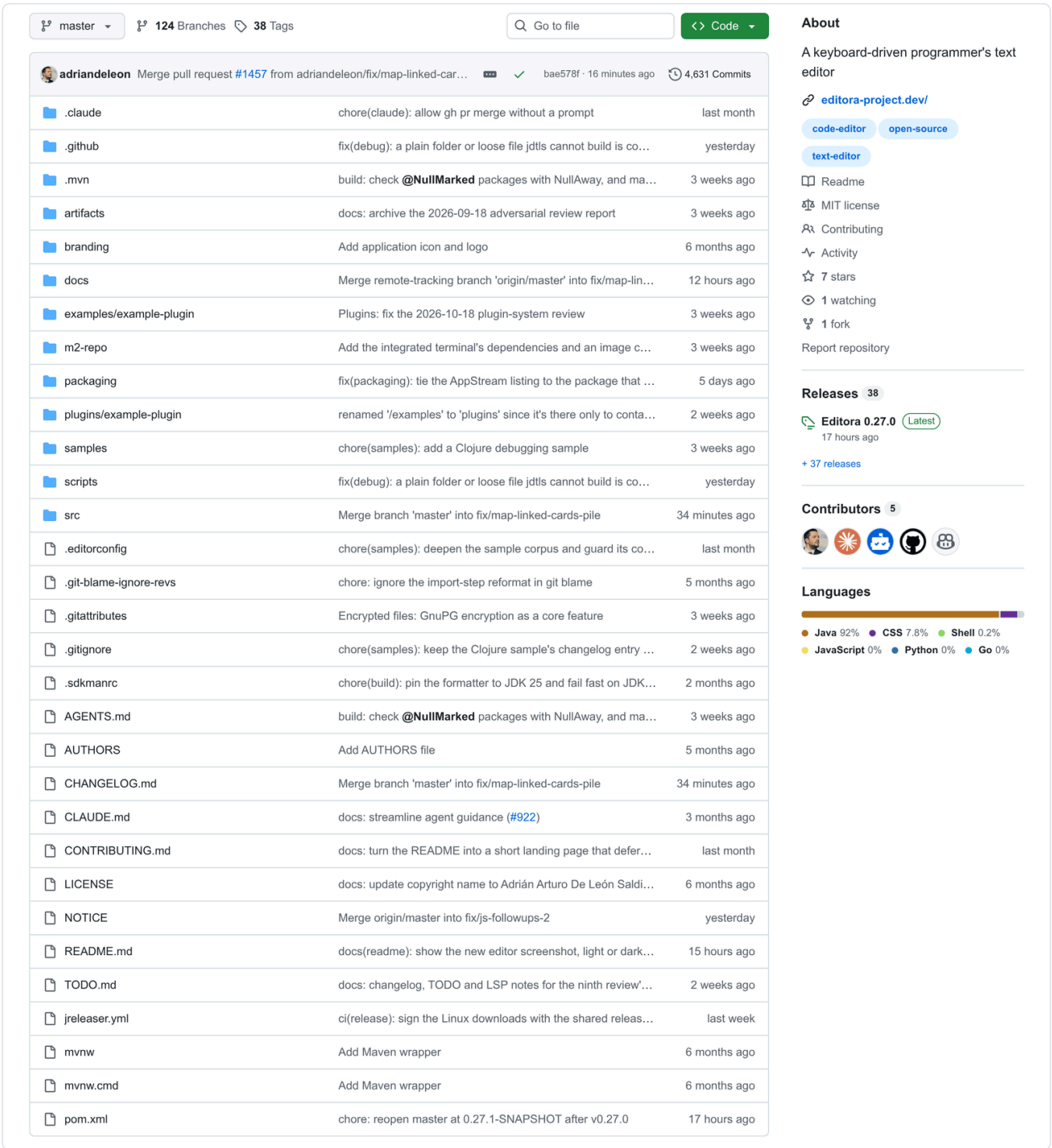
Everyday commands (identical in every repo)

```
./mvnw javafx:run                # run the app
./mvnw test                      # unit + headless FX tests, writes the JaCoCo report
./mvnw spotless:apply            # format
./mvnw verify                    # full gate: tests, coverage check, NullAway
./mvnw -Pfatjar package          # runnable fat jar for this OS
./mvnw clean -Pdist package      # native installer for this OS (clean is mandatory)
```

The `clean` in every `-Pdist` build is mandatory: incremental compilation can leave synthetic enum-switch classes out of `target/classes`, and jlink then ships an app whose keyboard fails on the first keypress.

Repository documents

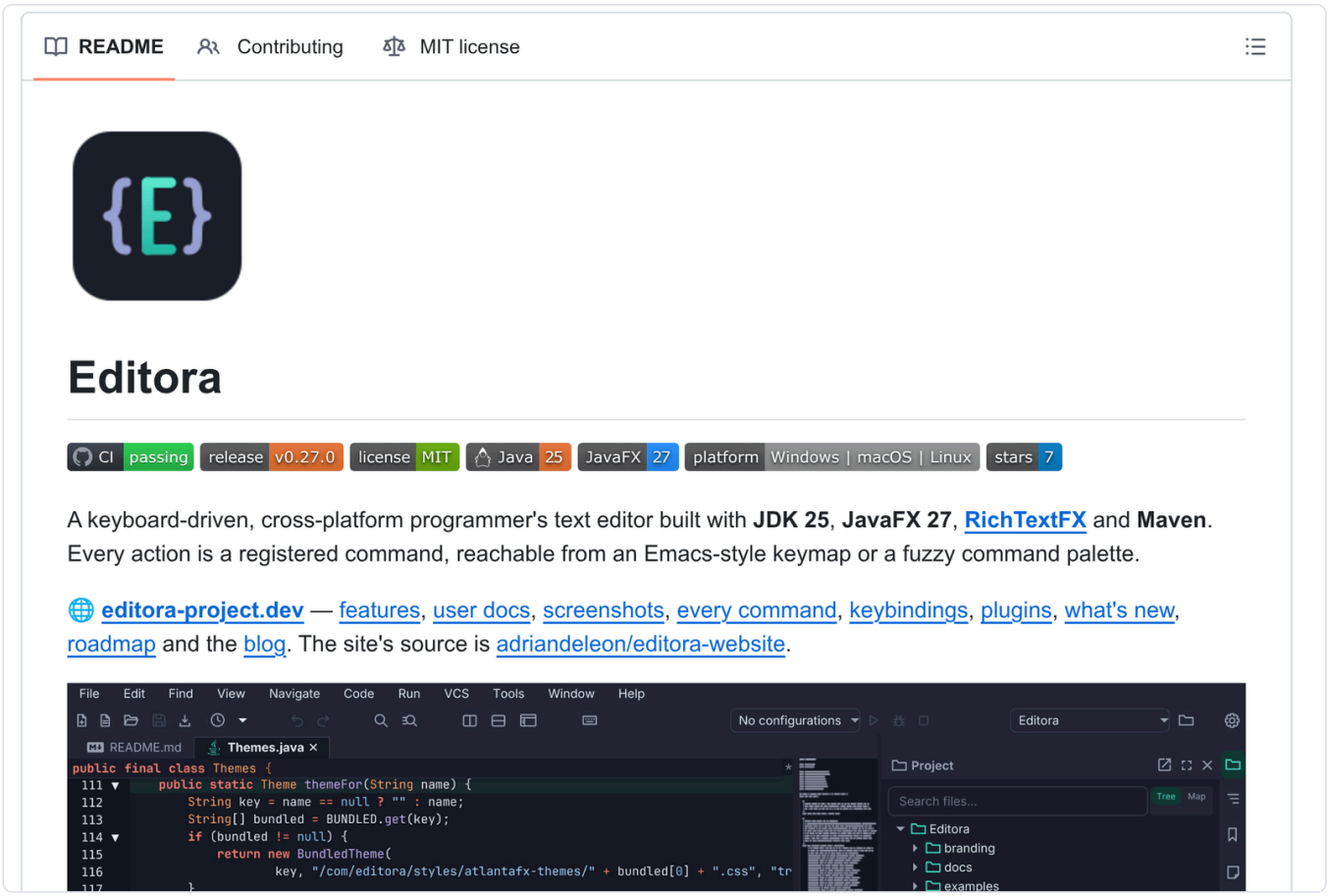
File	What it holds	Level
`README.md`	A landing page, about 200 lines: logo, badges, one-paragraph pitch, hero screenshot, highlights, install, build, command line, config folder, licence. Detail lives on the website.	`MUST`
`CHANGELOG.md`	Keep a Changelog 1.1 + SemVer. An `## [Unreleased]` section with Added / Changed / Fixed / Removed; each entry opens with a bold one-line summary written for users.	`MUST`
`LICENSE`	MIT, "Copyright (c) 2026 Adrián Arturo De León Saldivar".	`MUST`
`NOTICE`	App name, copyright, "licensed under the MIT License", then every bundled third-party component with its licence (Inter is OFL, AtlantaFX is MIT, OpenJFX is GPLv2+CPE…). Ship it inside the installers too.	`MUST`
`AGENTS.md` (+ a one-line `CLAUDE.md` pointing to it)	Operational rules for coding agents: essential commands, worktree-per-task, change checklist. Not a changelog.	`MUST`
`CONTRIBUTING.md`	How to build, test, format, and open a PR.	`MUST`
`docs/`	`architecture.md`, `conventions.md`, `testing.md`, `gotchas.md`, `release.md`, `building-and-packaging.md`, `decisions/` (ADRs).	`MUST`
`TODO.md`	The open list, kept in the same PR as the change.	`MAY`
`.github/ISSUE_TEMPLATE/bug_report.yml`	Issue form whose field ids (`environment`, `report`) the in-app Report a Problem fills.	`MUST`
`branding/`	Source SVG of the icon; generated PNG/ICNS/ICO live in `packaging/`.	`MUST`



Editora's repository root: every file on this page, with the homepage, Readme, MIT license and Contributing links GitHub derives from them.

README badges (in this order)

```
[![CI](https://github.com/adriandeleon/<App>/actions/workflows/ci.yml/badge.svg)](…/actions/workflow
[![Release](https://img.shields.io/github/v/release/adriandeleon/<App>)](…/releases/latest)
[![License: MIT](https://img.shields.io/github/license/adriandeleon/<App>)](LICENSE)
![Java](https://img.shields.io/badge/Java-25%2B-orange?logo=openjdk&logoColor=white)
![JavaFX](https://img.shields.io/badge/JavaFX-27%2B-1e90ff)
![Platforms](https://img.shields.io/badge/platform-Windows%20%7C%20macOS%20%7C%20Linux-lightgrey)
[![Stars](https://img.shields.io/github/stars/adriandeleon/<App>?style=flat)](…/stargazers)
```



The top of Editora's README: logo, badges in this order, a two-sentence pitch, links to the website, then the hero screenshot.

The hero screenshot is a `<picture>` with a dark `srcset` under `prefers-color-scheme: dark`, so GitHub shows the screenshot that matches the reader's theme. Screenshots are taken at 2× by a headless harness so they can be retaken after any UI change. The README states that the app is built with the help of AI coding tools.

Code quality gates

- **NullAway is an error, not a warning.** Put `@Nullable` on what may be null and fix the report; do not suppress it. A new package starts with its own `package-info.java` carrying `@NullMarked`.
- **Coverage floors are a regression net, not a target.** The build-wide floor is 0.90 lines and methods. Each package of 300+ lines gets its own floor about three points below its measured level; when a package's coverage rises, ratchet its floor up. A test (`BuildHygieneTest`) fails when the documented floor table and the pom disagree.
- **Prefer a pure decision helper plus unit tests.** Use the headless FX harness only when controller or toolkit behaviour is the point.
- **Tag FX tests** `@Tag("fx")`, so the pure suite runs alone with `-DexcludedGroups=fx`.
- **Guard tests that pay for themselves across the family:**
  - *Command sweep* runs every registered command in a real headless window (`CommandSweepFxTest`).
  - *Message parity*: every locale has every key; every `command.<id>` has a `.desc` (`MessagesTest`).
  - *Settings round trip*: every switch, number and choice survives a restart (`SettingsRoundTripFxTest`).
  - *Contrast*: text tokens keep their ratios in every theme (`StyleContrastTest`).
  - *Release supply chain*: signing and attestation steps stay where they are (`ReleaseSupplyChainTest`).
- **No sleeps in tests.** Own workers and FX callbacks with a scope (`AsyncTestScope`) and use latches.
- **A test window writes only inside its temp folders**, never in the real home directory.

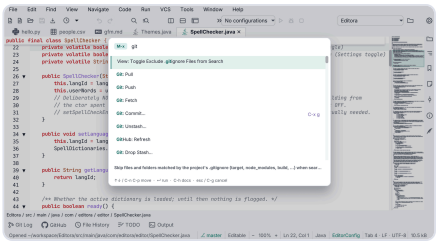
Keyboard first

Commands

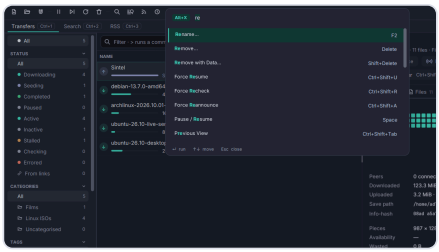
- Every user-facing action is a `Command` (id, title, runnable) registered in one `CommandRegistry`.
- Command ids are dotted and grouped by area: `file.open`, `view.toggleStatusBar`, `help.reportProblem`.
- Titles come from `command.<id>` and palette descriptions from `command.<id>.desc` in the message catalogs, so a registered command is localized and discoverable without extra work.
- A command that cannot run in the current context is shown greyed with a tooltip that says why, not hidden.

Command Palette

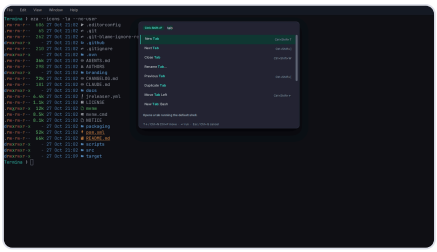
- **MUST.** Opened by `M-x` in Emacs keymaps and `Ctrl/Cmd+Shift+P` in the others.
- Fuzzy matching over title, id and description; recently used commands first; each row shows the active key binding.
- It is an **in-scene overlay** in the window's root `StackPane`, not a `javafx.stage.Popup`. On Windows a Popup does not reliably take OS keyboard focus and the whole app can stop receiving keystrokes.
- The same picker (`QuickOpen`) serves every other list choice: recent files, themes, keymaps, settings values.



Editora: M-x over the editor; typing git lists every Git command.



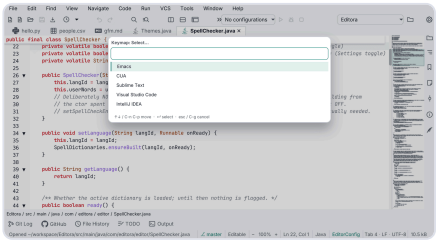
Torrenta: the same palette, filtered to torrent actions with their shortcuts.



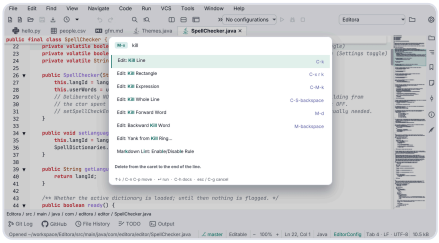
Termina: the palette over a shell session, filtered to tab commands.

Keymaps

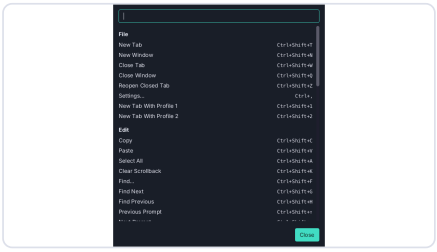
- Ship presets as JSON resources: Emacs, CUA, IntelliJ, VS Code and Sublime, with `.mac.json` variants where Cmd replaces Ctrl.
- Layer order: base keymap → user overrides → plugin keymaps → the user's own bindings again, so the user always has the last word (`KeymapLayers`).
- A keymap editor in Settings lists every command with its chord, searchable, with conflict warnings.
- Never bake a chord into translated text. Derive the label from the active keymap.
- Multi-key prefixes (`C-x …`) show a which-key style hint listing the possible next keys.



Keymap: Select… offers Emacs, CUA, Sublime Text, Visual Studio Code and IntelliJ IDEA.



M-x kill: each command shows its chord from the active keymap.

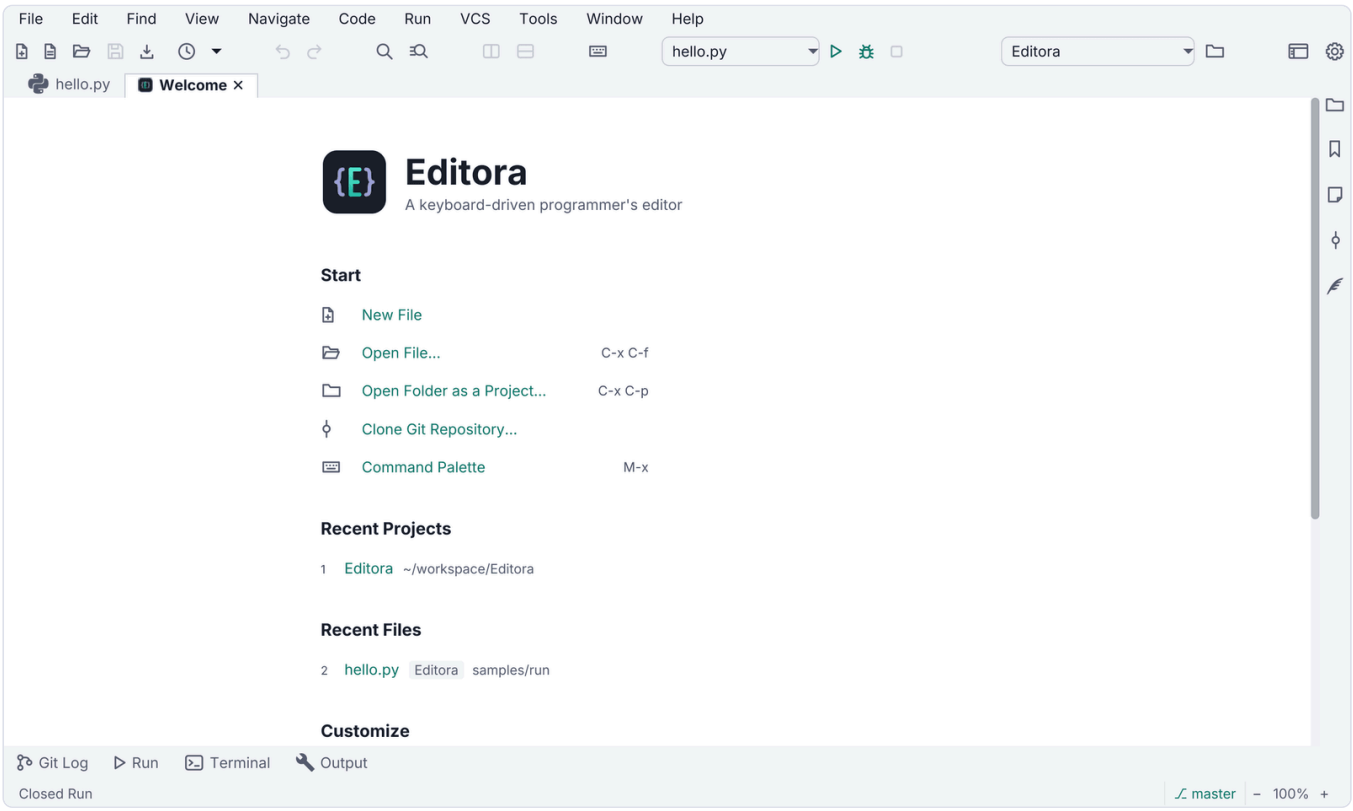


Termina's shortcuts reference: every key, grouped by menu.

Focus and accessibility

- Every interactive control is reachable with Tab and operable with Space/Enter.
- Keyboard focus is drawn by one `:focus-visible` block at the end of `app.css`.
- Icon-only buttons are built by a helper (`Icons.button`) that sets both the tooltip and the accessible text.
- On/off rows use a keyboard-operable switch (`SettingSwitch` , `TOGGLE_BUTTON` role). A bare AtlantaFX `ToggleSwitch` is mouse-only.
- Readable text uses `-color-fg-muted` or stronger; `-color-fg-subtle` (about 3:1) is for disabled or decorative text only.
- Rows wrap instead of truncating to "…". Check German at 1024 px wide.

Window shell

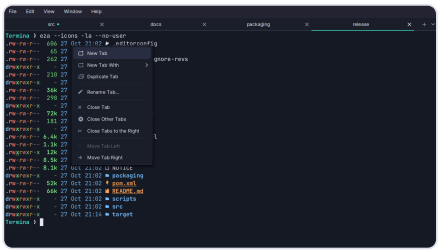


Editora's Welcome page: Start actions with their chords, Recent Projects, Recent Files and Customize.

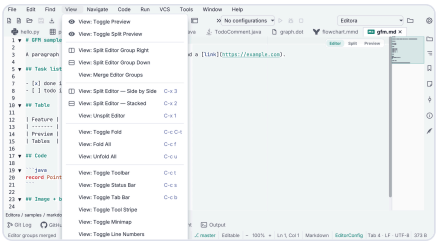
- **Status bar**, always present (toggleable). Its left side is the *echo area*: one line of feedback for the last action. Errors are red, warnings amber. Clicking it opens the Message Log.
- **Themes**: light and dark AtlantaFX themes plus a "follow the system" choice, switchable at run time without a restart. Semantic colours: amber = not durable yet (unsaved, read-only), red = broken, green = verified or added.
- **Title bar**: SHOULD use JavaFX 27's `HeaderBar` with `StageStyle.EXTENDED` to put the menu or tabs in the title bar, with a decorated-window fallback, and `useSystemMenuBar` on macOS.
- **Native dialogs** go through one helper (`Dialogs.styled`) because a dialog is its own scene and does not inherit the app stylesheet or the UI font.
- **Welcome page** when there is nothing to restore: Start actions (which run commands), Recent items, a Customize block (keymap and theme), Help links.
- **About** shows version, build, runtime (Java and JavaFX versions), licence, links, and the result of the last update check.
- **Window state**: size, position, maximized state and open items are restored on the next launch; a `--no-session` flag skips the restore.
- **Zoom**: UI and content zoom from the keyboard (`Ctrl/Cmd +` , `-` , `0`), persisted.
- **Modes** (MAY): Zen (no chrome), Simple (fewer menus for new users), Expert.



Mirada's HeaderBar title bar carries the file name and folder; the info panel and thumbnails are commands too.



Termina puts its tabs in the title bar with JavaFX 27's HeaderBar.

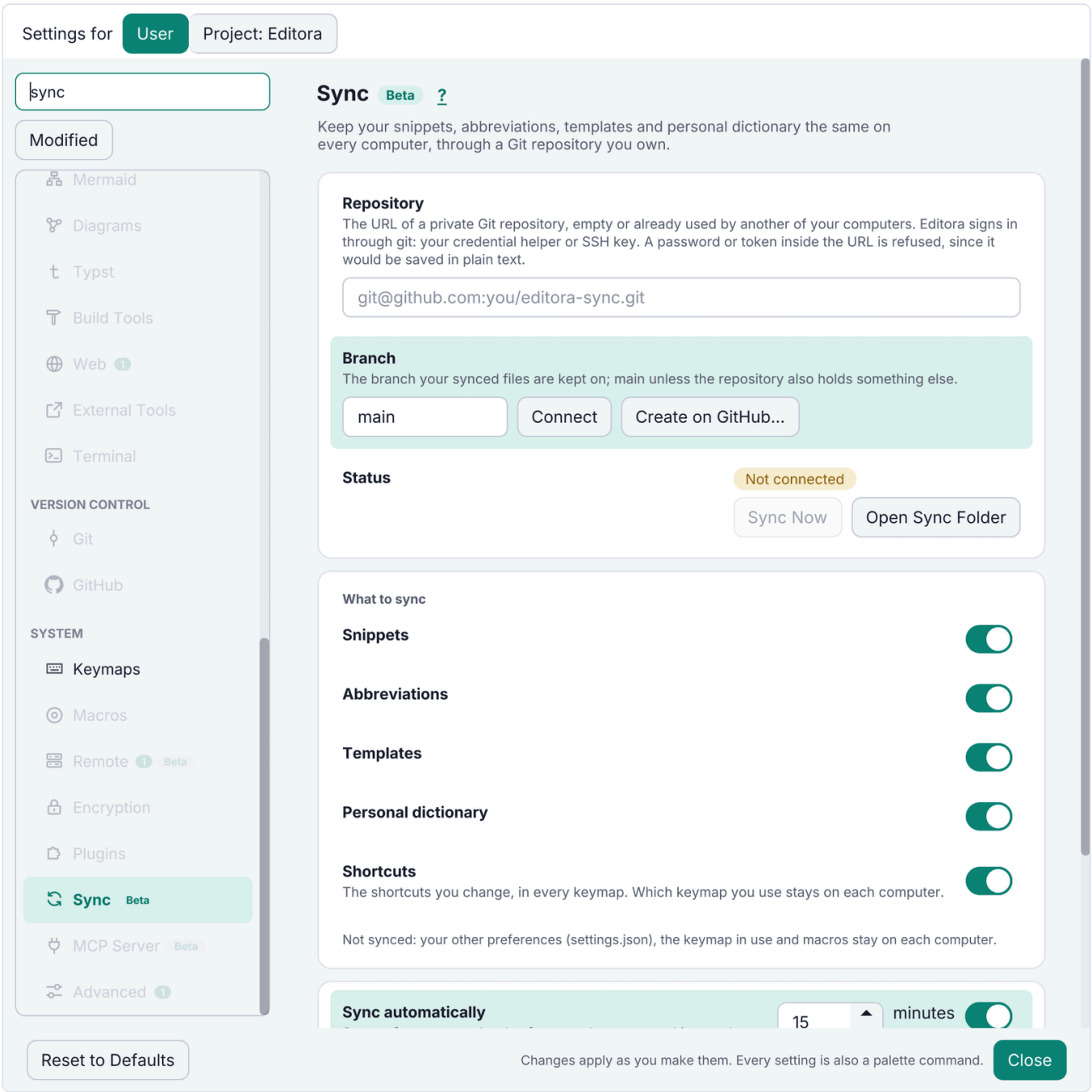


Every menu item is a command and shows its chord.

Settings

The Settings window

- **MUST.** Opened by `Ctrl/Cmd+,` and a palette command. A sidebar of pages with icons on the left, the page on the right, and a **search field on top**.
- **Search** finds a row by its title, description, visible text, its card or section heading, its page name and English keywords. Title hits rank above description hits; whole words above word starts. The page with the best match opens, and every non-matching row, heading and card is hidden (`SettingsSearch`).
- **Live apply.** No Save button. Text fields commit on Enter or focus loss; invalid numbers are marked and never committed.
- A change made in one window reaches every other open window.



Typing sync in Settings: pages without a match grey out and the best page opens.

Adding a setting touches five places

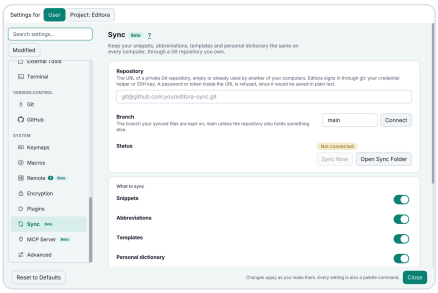
1. **Model** field, default and accessors, with a schema bump and migration.
2. **Control**, built once, writing the field and applying it.
3. **Row** on its page with search keywords.
4. **Load** line that sets the control from the model, safe to run at any time.
5. **Palette command** that flips or prompts the same field, plus strings in every catalog.

Config folder and schema

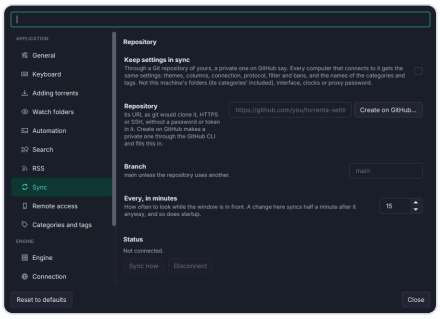
- Config lives in `~/.<app>` (`~/.<app>-dev` for `--dev`, any folder via `--config-dir` or `<APP>_CONFIG_DIR`).
- `settings.json` for preferences, `workspace-state.json` for session state, other stores in their own JSON files.
- Every versioned file has a `SCHEMA_VERSION` and a `v → v+1` migration step. A purely additive field uses an identity step.
- A file newer than the running build is backed up to `<name>.v<n>.bak` and defaults load: an older build never clobbers a newer config.
- **Export configuration** writes a timestamped `.zip` of the config folder (`ConfigExporter`). Import restores from one.

Settings Sync (Git / GitHub)

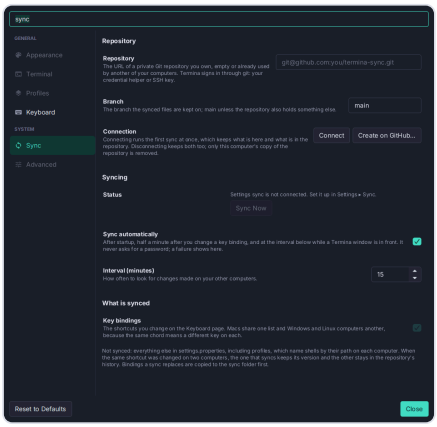
- **SHOULD.** Optional sync of settings, keybindings and user data through a Git repository the user owns.
- Setup by pasting a repo URL, or **Create on GitHub** through the `gh` CLI. The account that can read the repo is pinned per clone, so machines with two GitHub accounts work.
- Automatic (on change and at start) and manual triggers. Machine-local keys (window size, paths) never sync.
- Conflict rule: on a same-entry conflict *this machine wins*. A backup is taken before any sync that replaces local files.
- Status shows as a pill in Settings and as echo messages; failures go to the Message Log.
- Reference: `sync/SyncEngine` , `sync/SyncMerge` , `sync/SyncAccount` , `ui/SettingsSync` , `docs/subsystems/settings-sync.md` .



Editora's Sync page: repository, branch, Create on GitHub... and what to sync.



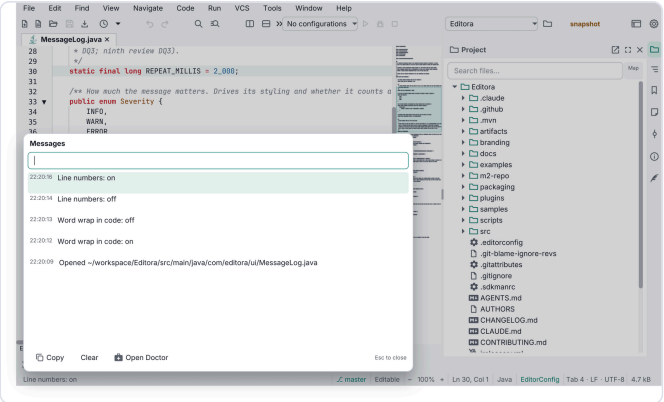
Torrenta's Sync page.



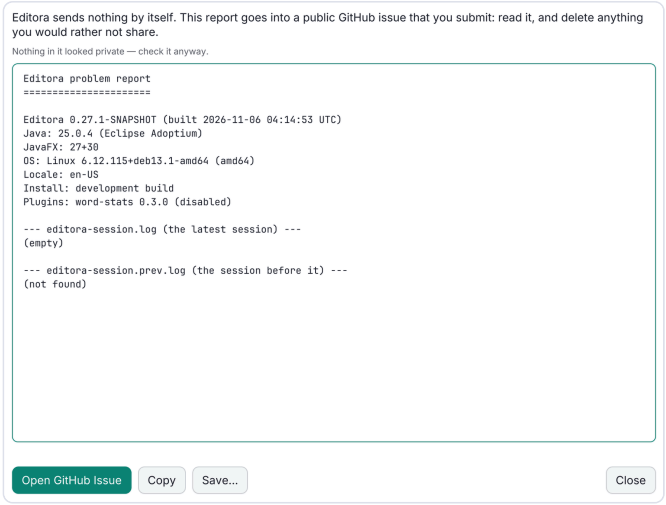
Termina's Sync page.

Diagnostics and support

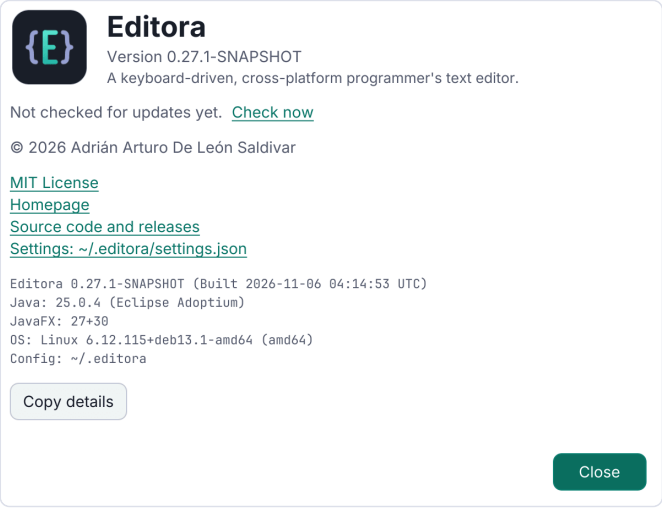
Feature	Behaviour	Level	Editora reference
Message Log	Session-only history of every echo-area message with `HH:mm:ss` times, newest first, bounded. Opened by clicking the status message; Esc closes; Clear empties it.	`MUST`	`ui/MessageLog`, `ui/MessageLogPopup`
Debug Log	Captures all `java.util.logging` records and uncaught exceptions in memory (bounded) **and** mirrors them to `<app>-session.log` in the config folder, keeping the previous one as `<app>-session.prev.log` so a crash is still visible next launch. A modeless window with Refresh, Copy, Clear, Export and Report a Problem. The only way to see logs in a packaged build.	`MUST`	`ui/DebugLog`, `ui/DebugLogWindow`
Uncaught exception handler	Installed first thing in `main`; routes to the Debug Log and an echo message. Never a stack trace dialog.	`MUST`	`DebugLog.install()`
Update notification	Reads GitHub `/releases/latest`. Once per app session (not per window), at most once a day, a few seconds after the first frame. A newer version shows a status-bar notice and a line in About; "Check for Updates" runs it by hand. Off switch in Settings. Never downloads or installs by itself.	`MUST`	`update/UpdateCheck`, `ui/UpdateCheckScheduler`, `ui/UpdateCoordinator`
Report a Problem	Help → Report a Problem builds a report (version, OS, runtime, settings summary, recent Debug Log), **redacts** it (home paths, user names, tokens, hosts), shows it editable, copies it to the clipboard and opens a prefilled GitHub issue from `bug_report.yml`. Over about 7,500 characters the link falls back to "paste from the clipboard". `<app> --report` does the same from the command line for when the app cannot start. Nothing is uploaded.	`MUST`	`report/ProblemReport`, `report/Redactor`, `report/ReportCommand`, `ui/ReportProblemWindow`
Doctor	A page that checks the environment (external tools, versions, permissions, paths) with a status per row and a fix or link for each failure.	`SHOULD`	`doctor/*`, `ui/DoctorPane`
Crash recovery	Unsaved user work is snapshotted off the FX thread and offered back after a crash.	`SHOULD` (apps with user documents)	`recovery/RecoveryService`



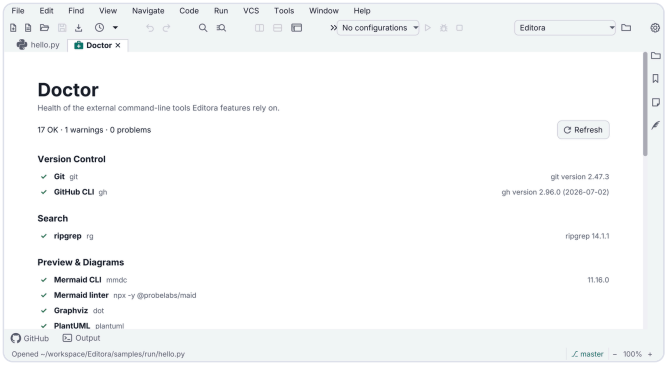
Clicking the status message opens the Message Log: this session's messages, newest first.



Report a Problem: a redacted, editable report that opens as a prefilled GitHub issue.



About: version, runtime, licence, links and the update status.



Doctor: each external tool with its version, or what to install.

Command line and launch

Every app accepts at least:

Flag	Meaning
`--help`, `--version`	Printed before the GUI starts; plain English, not translated.
`--config-dir <dir>`, `--dev`	Use another config folder (`~/.<app>-dev` for `--dev`).
`--new-instance`	Start a second process instead of handing off to the running one.
`--no-session`	Do not restore the last session.
`--report`	Write a redacted problem report and open the GitHub issue.
`[files…]`	Open these items.

- **Single instance per config folder.** A second launch hands its arguments to the running app over a local endpoint and exits (`ipc/SingleInstance`). On macOS, `openFiles` Apple Events reach the running app (`MacOpenFiles`). An advisory lock tells two processes on one folder about each other (`config/InstanceLock`).
- The headless-AWT guard is the first statement of `main` .
- A `.app` launched from Finder inherits a stripped `PATH` : resolve external tools through the login shell's `PATH` .

Internationalization

Languages

Every app ships **at least these six**. They are a floor, not the set: an app SHOULD add languages as people ask for them, and a new app can start with more.

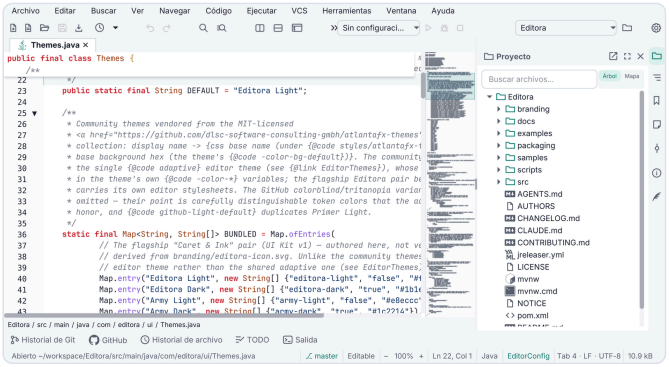
Language	Catalog
English (base)	`messages.properties`
Spanish	`messages_es.properties`
Portuguese	`messages_pt.properties`
French	`messages_fr.properties`
Italian	`messages_it.properties`
German	`messages_de.properties`

Adding a language:

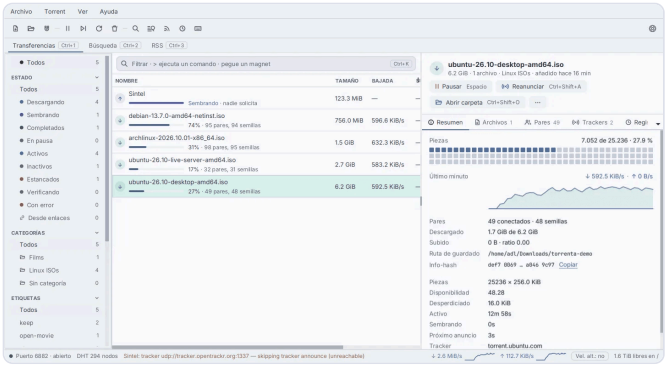
- Add `messages_<code>.properties` with every key, and the language to the Settings language picker. The parity test then holds it to the same keys as the others.
- Translate for real; do not ship English values in another language's catalog.
- Check the layout in the new language at 1024 px wide. German is today's longest; rows must wrap, never cut.
- Scripts beyond Latin need more than strings. Chinese, Japanese and Korean need a font fallback, because Inter has no CJK glyphs. Arabic and Hebrew need `NodeOrientation.RIGHT_TO_LEFT` on every scene.

Rules

- Never pass a raw English literal to a control: `tr("some.key")` from one `Messages` class.
- Every language has a real translation of every key. A test fails the build when a key is missing in any catalog.
- `MessageFormat` rules: in a value with a placeholder, double literal apostrophes; identifiers such as line numbers and ports use `{0,number,#}` so they are not digit-grouped.
- Untranslated on purpose: technical identifiers, `--help` and `--version` text.



Editora in Spanish: menus, tool windows and the status bar all come from the catalogs.



Torrenta in Spanish.

Packaging, signing and release

Targets

Each target builds on its own GitHub-hosted runner. jpackage and JavaFX natives are host-specific, so there is no cross-building.

Target	Runner	Downloads
Linux x64	`ubuntu-latest`	`.deb`, `.rpm`, `.AppImage`, `.tar.gz`, fat jar
Linux arm64	`ubuntu-24.04-arm`	`.tar.gz` (and `.deb` where the app's tooling allows), fat jar
macOS arm64	`macos-15`	`.dmg`
macOS x64	`macos-15-intel`	`.dmg` (the last Intel image, good to about Aug 2027)
Windows x64	`windows-latest`	`.msi`, fat jar
Windows arm64	○	**Not possible yet**: OpenJFX publishes no `win-aarch64` natives (JDK-8314064). ARM users run the x64 MSI under emulation.

- Downloads are named `<App>-<version>-<target>.<ext>`.
- jpackage builds the installers through the Maven `dist` profile. **JReleaser** (`jreleaser.yml`) only creates the GitHub release: it attaches the files, `checksums_sha256.txt` and the changelog section.
- The maven version is the single source of truth: `-SNAPSHOT` on `master`, a tag `vX.Y.Z` releases, a post-release job bumps to the next snapshot.
- GitHub releases are immutable, so a **release-asset check** fails the run before publishing when an expected file is missing.
- **Linux desktop integration**: a `.desktop` file whose `StartupWMClass` is the JavaFX app class, plus AppStream metainfo `io.github.adriandeleon.<App>`.
- **AOT cache** (JDK 25 Leyden): a training run during packaging cuts start-up time; Linux legs run it under `xvfb`.

Install methods

Channel	Platform	What the user runs	How it is published	Level
GitHub release	All	Download the `.dmg`, `.msi`, `.deb`, `.rpm`, `.AppImage` or `.tar.gz`	JReleaser attaches every file, with checksums	MUST
Homebrew cask	macOS (Apple Silicon)	`brew install --cask adriandeleon/tap/<app>`	JReleaser's `brew` packager writes the cask to `adriandeleon/homebrew-tap`	SHOULD
winget	Windows x64	`winget install AdrianDeLeon.<App>`	JReleaser's `winget` packager opens a pull request on `microsoft/winget-pkgs` from the `adriandeleon` fork	SHOULD

- **One tap for the family.** The official homebrew-cask repository has notability rules a new project does not meet, so the casks live in a personal tap. A cask takes exactly one artifact: it serves Apple Silicon, and Intel Macs use the `macos-x64` DMG.
- **winget is reviewed by Microsoft.** The first submission for a new publisher is checked by a person, not merged automatically. The package identifier is `AdrianDeLeon.<App>`, the moniker the app's lower-case name.

- **Both need a `PACKAGES_TOKEN` secret**, a classic personal access token with `repo` scope: the workflow's own token cannot write to the tap or open a pull request on the fork. Without it JReleaser skips both packagers and the GitHub release still publishes.
- `jreleaser package --basedir .` writes the cask and the winget manifests locally, to check them without releasing.
- **Status on 2026-11-06:** the tap exists and holds no casks yet. Torrenta and Termina have both packagers configured, waiting for the token. Mirada's first winget submission (microsoft/winget-pkgs#471599) is open.

Signing

Platform	How	Status in the family
macOS `.dmg`	Developer ID Application certificate (team 8JKSF797L8), hardened runtime with per-app entitlements, `notarytool` submission, stapled ticket, `spctl` check, then a smoke launch of the notarized app. Secrets per repo: `MACOS_CERT_P12`, `MACOS_CERT_PASSWORD`, `APPLE_API_KEY_P8`, `APPLE_API_KEY_ID`, `APPLE_API_ISSUER`. A tag without the secrets fails.	**Done** in all six
Linux `.deb` `.rpm` `.AppImage` `.tar.gz`	One shared OpenPGP release key ("Adrian De Leon (release signing)"). A separate `sign-linux` job: `rpmsign` for `.rpm`, a debsigs origin signature for `.deb`, and an armoured `.asc` beside every file. Public key committed as `packaging/linux/signing-key.asc` and attached to each release. Secrets `LINUX_GPG_KEY`, `LINUX_GPG_PASSPHRASE`. Shared scripts `sign-linux-packages.sh` and `check-linux-signatures.sh`.	Editora, Torrenta, Termina, Modula; Mirada partly; nux not yet
Windows `.msi`	Authenticode. **Open decision**: SignPath Foundation (free for OSS), Certum Open Source cert, or Azure Trusted Signing. Until then MSIs are unsigned and SmartScreen warns.	**Not done** in any app
All	GitHub **build provenance attestations** (`actions/attest-build-provenance`) for every download; `gh attestation verify` checks them.	Editora, Termina

- Signing steps skip cleanly when the secrets are absent, so a fork or a dry run still builds.
- The certificate renews before 2027-11-30; the Apple membership before 2027-10-31.

CI/CD (GitHub Actions)

Workflow	When	What	Level
`ci.yml`	Every push to `master` and every pull request	**Format** (`spotless:check`) and **Build & Test** (`./mvnw verify`, coverage gate included) on JDK 25. Both are required status checks for `master`.	MUST
`full-tests.yml`	Daily (cron), on a pull request labelled `full-ci`, and before every release	The full suite on Linux, macOS and Windows; flaky tests listed in the job summary.	SHOULD
`release.yml`	A `v*` tag that matches the pom version; `workflow_dispatch` as a dry run	The five-target matrix, signing jobs, attestations, the release-asset check, JReleaser, then the bump to the next snapshot. A dry run builds and signs and publishes nothing.	MUST

Every workflow:

- Pins each third-party action to a commit SHA, with its version in a comment (`actions/setup-java@b843ebf… # v6.1.0`). Dependabot updates `github-actions` and `maven` weekly.
- Starts from `permissions: contents: read`; only a job that publishes asks for write access, in its own block.
- Sets up Temurin JDK 25 with `actions/setup-java` and `cache: maven`.
- Puts `timeout-minutes` on every job, and uses `concurrency` to cancel a superseded pull-request run, never a run on `master`.
- Uploads the surefire reports when tests fail and the JaCoCo report always; writes the coverage totals to the job summary.

- Skips signing cleanly when the secrets are absent (forks, dry runs); a tag without them fails.
- Launches the packaged app once on each OS before a release (a smoke test).

Feature matrix

Surveyed on 2026-11-05 from each repository's default branch. ● has it, ◐ partly or differently, — not found.

What the apps are.

App	What it is	Homepage	Source
Editora	Programmer's text editor (the reference)	editora-project.dev	adriandeleon/Editora
Torrenta	BitTorrent client	torrenta-project.dev	adriandeleon/torrenta
Termina	Terminal emulator	termina-project.dev	adriandeleon/Termina
Modula	Software-defined-radio receiver	○	adriandeleon/Modula
Mirada	Fast image viewer (GraalVM native image)	○	adriandeleon/Mirada
nux	UPS monitor for Network UPS Tools, in the tray	○	adriandeleon/nux

Platform and repository

	Editora	Torrenta	Termina	Modula	Mirada	nux
JDK 25+ and JavaFX 27+, JPMS module, `mvnw`	●	●	●	●	●	●
AtlantaFX 3	●	●	◐ vendored CSS	● Primer	● Cupertino	● Primer
Inter UI font (SHOULD)	●	●	●	○ Plex Mono	○	○
NullAway + every package `@NullMarked`	●	●	●	●	●	○
Spotless (Palantir)	●	●	●	●	●	○
JaCoCo gate	● 0.90 L/M	● 0.90 L	◐ per-package	○	○	○
Headless FX tests	●	●	●	○	●	●
Six or more message catalogs	●	●	●	○	○	●
README badges + screenshot	●	●	●	●	◐ no shot	●
Keep a Changelog format	●	◐	◐	●	◐	●
MIT `LICENSE` + `NOTICE`	●	●	●	●	●	●
`AGENTS.md` canonical, `CLAUDE.md` a pointer	●	◐ `CLAUDE.md` only	●	◐ `CLAUDE.md` only	●	◐ copy

	Editora	Torrenta	Termina	Modula	Mirada	nux
`CONTRIBUTING.md`	●	○	○	○	○	○
`docs/` and `branding/`	●	●	●	●	●	●
`bug_report.yml` issue form	●	●	○	○	○	○
Dependabot	●	●	○	●	○	●

Release and signing

	Editora	Torrenta	Termina	Modula	Mirada	nux
JReleaser	●	●	●	○	○	○
Five-target matrix (Linux x64/arm64, macOS arm64/x64, Windows x64)	●	◐	●	◐ no Linux arm64	●	●
macOS signed + notarized `.dmg`	●	●	●	●	●	●
Linux OpenPGP `.asc` signatures	●	●	●	●	◐ rpm + checksums	○
Windows `.msi` Authenticode	○	○	○	○	○	○
Build provenance attestations	●	○	●	○	○	○
Homebrew cask / winget manifests	○	●	●	○	◐ winget	○
`.desktop` file + AppStream metainfo	●	●	●	●	◐ no metainfo	◐ no metainfo
AOT start-up training	●	○	●	○	●	○

User-facing features

	Editora	Torrenta	Termina	Modula	Mirada	nux
Command registry + Command Palette (in-scene overlay)	●	●	●	●	●	●
Keymap presets	● 9	○	○	○	○	○
Keybinding editor / shortcuts reference	●	●	●	○	◐	○
Settings window	●	●	●	●	●	●
Settings search	●	●	●	○	○	○
Settings sync (Git / GitHub via `gh`)	●	●	●	○	○	○

	Editora	Torrenta	Termina	Modula	Mirada	nux
Message Log (status-bar history)	●	●	○	○	○	○
Debug Log + uncaught exception handler	●	●	●	○	○	○
Update notification (GitHub releases)	●	●	●	●	○	●
Report a Problem → GitHub issue (+ `--report`)	●	●	○	○	○	○
About window with runtime versions	●	●	●	●	●	●
Welcome page	●	●	○	○	○	○
Doctor / diagnosis	●	◐ per torrent	○	◐ native libs	○	○
Single instance (hand off to running app)	●	●	○	○	○	●
`--config-dir` and other CLI flags	●	●	●	○	○	◐
Light / dark / follow system, live switch	●	●	●	●	●	●
Extended title bar (`HeaderBar`, `StageStyle.EXTENDED`)	●	◐ opt-in	●	○	●	○
Desktop notifications	●	●	●	●	○	●
Tray icon (SNI over D-Bus, AWT fallback)	○	●	○	●	○	●
Window geometry restore	●	●	●	○	●	○
Config export / import	●	○	○	○	○	○
Plugins	●	○	○	○	○	○

Shared features

Beyond the list this guide started from, these already exist in two or more apps and belong in the guideline:

1. **`AppInfo` + `build-info.properties`.** Version, build time and commit come from a Maven-filtered resource, read by About, `--version`, the update check and problem reports. *(all six)*
2. **Shared `update/` package** (`UpdateCheck` pure parser, `UpdateService`, `ReleaseInfo`). *(5 apps)*
3. **Pure palette core shared verbatim**: `FuzzyMatch`, `MatchText`, `KeyChord`, `PaletteKeys`, `TextInputKeys`. Emacs-style caret keys work inside the palette's query field. *(Editora → Termina → Mirada, Torrenta)*
4. **Shortcuts reference window**: a searchable list of every key, opened from Help. *(Termina, Mirada)*
5. **Extended title bar** with JavaFX 27 `HeaderBar` and `StageStyle.EXTENDED`, falling back to a decorated window; the macOS system menu bar via `useSystemMenuBar`. *(Editora, Termina, Mirada, Torrenta opt-in)*
6. **macOS app-menu integration**: About and Settings in the application menu, `openFiles` Apple Events reaching the running app (`MacOpenFiles` / `MacHandlers`). *(Editora, Mirada, Torrenta)*
7. **Tray package** (`AwtTray`, `SniTray`, `StatusNotifierItem` over dbus-java) plus **autostart** (XDG autostart, LaunchAgent, `HKCU\…\Run`) and `--minimized`. *(Torrenta, Modula, nux)*
8. **Desktop notifications** through one `Notifier`: `notify-send` / `gdbus` on Linux, `osascript` on macOS, a PowerShell toast on Windows, the tray as fallback. *(Torrenta, Termina, Modula, nux, Editora)*
9. **Window geometry that stays on screen**: `WindowGeometry` / `WindowPlacement` restore size and position but move a window back when its monitor is gone. *(Editora, Torrenta, Termina, Mirada)*
10. **Linux desktop integration**: a `.desktop` file whose `StartupWMClass` matches the JavaFX app class (`com.<app>.App`), AppStream metainfo `io.github.adriandeleon.<App>`, multi-size PNG icons.
11. **Package-manager channels**: Homebrew cask and winget manifests generated by JReleaser. *(Torrenta, Termina, Mirada)*
12. **Atomic config writes** with a file lock, and values clamped to a valid range when read. *(Termina, nux, Mirada)*
13. **Dev capture / screenshot mode** (`DevCapture`, nux `--screenshot=`): renders the real UI headless at 2× for README, website and bug-report images. *(Torrenta, Termina, nux; Editora uses an external headless harness)*
14. **Smoke launch in CI**: the packaged app is started and checked (still running after 20 s, or `--version`) on every OS before release. *(all six)*
15. **Release dry run**: `workflow_dispatch` builds and signs with throwaway credentials and publishes nothing. *(Editora, Modula, Mirada, nux, Termina)*
16. `SECURITY.md` with how to report a vulnerability privately. *(Modula; worth adding everywhere)*
17. **Simple mode / density**: fewer menus and a compact layout for new users. *(Editora, Torrenta)*
18. **Zoom levels** stored per window kind, `Ctrl/Cmd +/-/0`. *(Editora, Termina)*
19. **Undo for destructive list actions** (remove a torrent, close a tab) and **reopen closed tab**. *(Torrenta `UndoStack`, Termina and Editora `ClosedTabs`)*

Gaps per app

App	Missing MUSTs
Editora	Windows MSI signing.
Torrenta	Keep-a-Changelog format; `AGENTS.md` (only `CLAUDE.md` is committed); `CONTRIBUTING.md`; Windows signing; attestations.
Termina	Message Log; Report a Problem; single instance; `bug_report.yml`; `CONTRIBUTING.md`; full coverage gate; Windows signing.
Modula	JaCoCo; headless FX tests; i18n; Settings search and sync; Message/Debug Log; Report a Problem; `AGENTS.md`; `CONTRIBUTING.md`; `bug_report.yml`; JReleaser; Linux arm64.
Mirada	JaCoCo; i18n; Settings search and sync; Message/Debug Log; uncaught handler; update check; Report a Problem; `CONTRIBUTING.md`; `bug_report.yml`; JReleaser.
nux	NullAway; Spotless; JaCoCo; Settings search and sync; Message/Debug Log; Report a Problem; `CONTRIBUTING.md`; `bug_report.yml`; Linux signatures; JReleaser.

Web version: adriandeleon.github.io/adl-desktop-apps-guidelines. Source: github.com/adriandeleon/adl-desktop-apps-guidelines. Version 1.0.0, 2026-11-06.